Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic · 36 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic · 24 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic · 35 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic · 35 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic · 36 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic · 35 trials
Total training paths	10 · 341 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	Accel
Sensor columns	78 channels
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats → feature dim 624
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=1360 · test=69

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	1.0000	0.0000	0.9855

Per-class hold-out F1

Class	Random Forest	Support
Double_Lower	1.000	10
Double_Nothing	0.941	9
Double_Pistol_Recoil	0.952	10
Double_Raise	1.000	10
Double_Wiggle	1.000	10
Double_cmere	1.000	11
Drum_Roll	1.000	9
macro avg	**0.985**	**69**

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.9143

Per-class external accuracy

Class	Random Forest	Support
Double_Lower	1.000	5
Double_Nothing	0.400	5
Double_Pistol_Recoil	1.000	5
Double_Raise	1.000	5
Double_Wiggle	1.000	5

Double_cmere	1.000	5
Drum_Roll	1.000	5